

TrES-2b

R-2413

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_110626 · created 2026-08-21T22:48:53+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2455738.947672 +/- 8.8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1462 +/- 0.0088
Transit depth (Rp/Rs)^2	2.14 +/- 0.26 [%]
Orbital Inclination (inc)	83.96 +/- 0.43
Airmass coefficient 1 (a1)	0.9367 +/- 0.0072
Airmass coefficient 2 (a2)	0.059 +/- 0.019
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	#2 - [86, 240]
Optimal Aperture	4.41
Optimal Annulus	6.12
Transit Duration (day)	0.0736 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1462 $\pm$ 0.0088 vs 0.1254 (deviation 2.19 σ)
Duration fit vs published	0.0736 d vs 0.0746 d (deviation 0.26 σ)
Mid-time O-C	-0.1 min
Depth SNR	15.4
Residual scatter	0.77 %

Light curve

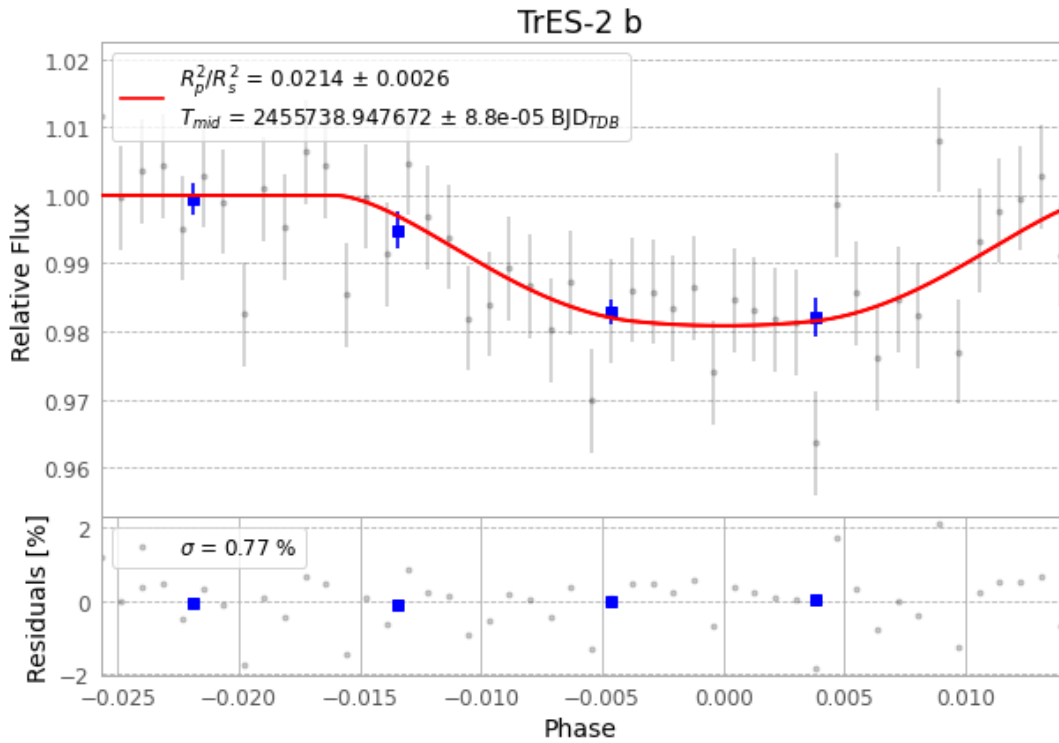


Figure 1 — FinalLightCurve_TrES-2 b_2011-06-26.png (SHA-256 86002f05b003d8d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e150c6342646d083...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), TRES-2110626090913.FITS → TRES-2110626120013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-110626.log (9a7973caedb1...), FinalLightCurve_TrES-2 b_2011-06-26.png (86002f05b003...), FinalLightCurve_TrES-2 b_2011-06-26.csv (29cf61ede749...)

Compute resources

Wall time 164.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 22:48Z.